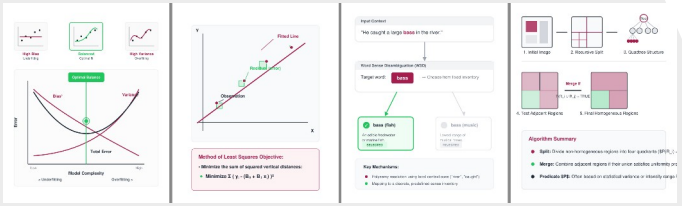
IPD PROJECT REVIEW 2026

TITLE PAGE

- **Project – Learnova**
- **Theme – EdTech · Generative AI**
- **Category – Software**
- **Group No. – 1 (AI & DS, DJSCE)**
- **Team – Swayam · Kawaljeet · Jeevesh · Rishit**
- **HoD – Dr. Pratik Kanani**



Learnova

LEARNOVA — IDEA

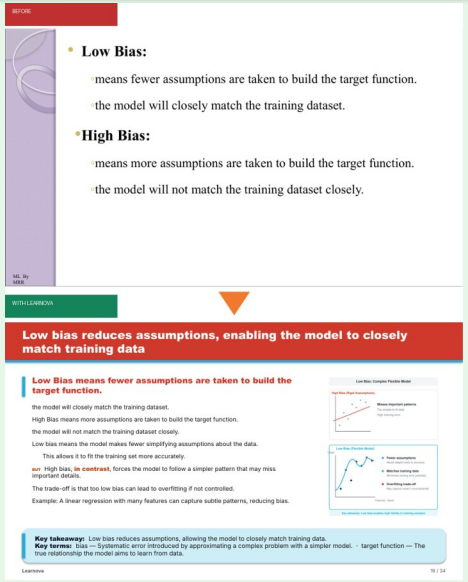
5 real lecture decks: 83 untitled slides · 0 diagrams · 0 quizzes

Proposed Solution: Teachers reuse old decks for years and students revise from them before every exam, yet most are walls of bullets with pictures of equations. Learnova takes the deck a teacher already has (PPTX, PDF, Word notes or a topic) and rebuilds it as an editable PowerPoint, an interactive web deck and a study pack — every point kept, plus a visual, an illustration, quizzes, solved steps and labs.

Detailed explanation of the proposed solution

- **Slide Reader:** every page read with titles, bullet levels, tables, figures; OCR only for picture slides.
- **Decision Engine:** 27 detectors pick process, timeline, comparison, definition, chart or cards — with a reason.
- **Assertion Titles:** each title states the point of its slide; split parts are named, never "(2/5)".
- **Illustrations:** AI-drawn diagram, teacher's figure or offline scene on every slide, linked as an HTML page.
- **Explainers & Labs:** beginner step-by-step solutions and hands-on labs for hard topics.
- **Outputs:** editable PPTX, web deck, study pack, share link, tutor.

How it addresses the problem



Innovation and uniqueness of the solution

- **Redesign, not generate:** starts from the teacher's own deck; other tools start from a prompt.
- **Faithful by design:** source text kept verbatim; 98–100% of facts retained on real decks.
- **Explainable choices:** every visual decision shows its rule and confidence.
- **PSF metric + CLASS:** our own slide-fitness score and load-optimal slide splitting.
- **Two safety nets:** 5 Gemini keys rotate; if all fail, offline rules finish the deck.
- **Verified AI output:** diagrams sanitised, labs code-checked, titles grounded in the slide.
- **Works offline:** no paid licence, runs on a college laptop.

PHASE 1 · Sep–Dec 2025

Research and design

- Literature + competitor study
- Problem, architecture, UI plan

PHASE 2 · Jan–Jul 2026

Build the platform

- Prototype V1 → Learnova V2
- Engine, builders, accounts

PHASE 3 · Aug–Oct 2026

Real decks, AI, research

- Reader + OCR, illustrations, labs
- AI key pool, papers, review

LIVE CODE

github.com/SDP42/LearnovaV2

Open source · 755 tests · demo decks

Learnova

TECHNICAL APPROACH

Methodology and process for implementation

INPUTS

PowerPoint .pptx
teacher's old deck, any template

Slide PDF
exported or scanned lecture slides

Word / notes
.docx, markdown or plain text

Typed topic
a syllabus line or a prompt

Teacher labs
own HTML labs added via API

INGEST

LibreOffice convert
PPTX → PDF, cached by file hash

Slide reader
titles, bullet levels, tables, figures

OCR decision
Tesseract only when γ ≤ 0.42, conf ≥ 62

Repair rules
logos, page numbers, glued titles

Colour DNA
source palette becomes the theme

ENGINES

Planner + CLASS
load-optimal split, agenda, glossary

Decision engine
27 detectors → visual + reason

Layout engine
glyph-measured, 11 pt floor, no breaks

Explainer solvers
exact steps with reasons (SymPy)

PSF scorer + audit
fitness, overflow, fact retention

AI SERVICE

LLM router
Gemini ×5 → Groq → NVIDIA

AI polish
assertion titles, plain explanations

Topic diagrams
SVG sanitised → PNG + HTML page

AI labs
JS checked, no network access

Offline fallback
rules, scenes, structure labs

OUTPUTS

Editable PPTX
native shapes, linked pictures, notes

Web deck
click reveals, labs, flip cards

Study pack
flashcards, Anki, quiz bank, cheat sheet

Share + present
view-only link, presenter console

Deck tutor
answers only from the deck

1 · Read faithfully

Every page, title, level, table and figure recovered; nothing summarised away

2 · Decide with rules

27 detectors + PSF score: 97% right visual on 101 gold slides, < 1 ms

3 · Enrich with AI

Titles, diagrams, labs from 5 Gemini keys — each checked before use

4 · Build and audit

PPTX + web deck; 0 broken words, 98–100% facts kept, 755 tests

Technologies to be used

Frontend: React 18 · Vite · Tailwind CSS · Reveal.js web deck · PWA

Backend: Python 3.12 · FastAPI · Pydantic · job queue · SQLite · OTP auth

Documents: python-pptx · PyMuPDF · LibreOffice · python-docx · Tesseract OCR

AI / NLP: Gemini (5-key pool) · Groq · NVIDIA NIM via failover router

Maths & labs: SymPy exact solvers · JavaScript lab engine · SVG

Quality & DevOps: pytest (755 tests) · layout audit · GitHub · Docker

WORKING PROTOTYPE · real screens from Learnova



Learnova

FEASIBILITY AND VIABILITY

Running today · pilot-ready · every risk has a safeguard

Analysis of the feasibility of the idea

TECHNICAL — 755 tests
Working end to end: FastAPI back end, React front end, 755 automated tests and a layout audit; each stage is a separate module.

DATA — 7 real decks
Input is the college's own material (PPTX, PDF, DOCX). Tested on 7 real decks of 15–216 slides from NLP and ML courses.

ECONOMIC — ₹0 offline
Free offline; Gemini free-tier keys optional (pool of 5). A department server costs less than one paid AI-slides licence.

OPERATIONAL — editable
Teachers stay in control: every slide editable, every automatic choice shows its reason; students need only a browser.

SECURITY — sanitised
OTP sign-in, hashed codes, rate limits; AI SVG and lab code sanitised and checked; share links are view-only.

SCALE — ~35 s / 216 pp
216-page PDF → full deck in ~35 s offline; work runs in parallel and is cached by content hash.

Potential challenges and risks



Strategies for overcoming these challenges

1. **Faithful Reader** — vs Messy PDFs
Page reader, title DP split and OCR decision keep messy PDFs, logos and picture slides correct.

2. **Grounded AI** — vs Hallucination
AI titles and explanations accepted only if they reuse the slide's words; numbers come from SymPy.

3. **Key Pool + Offline** — vs Quota / cost
5 Gemini keys, per-key pacing and rotation; Groq/NVIDIA failover; offline rules always finish.

4. **Glass-box Decisions** — vs Black-box
Every visual shows its rule, confidence and alternatives; slide stays editable in the editor.

5. **Measured Layout** — vs Broken layout
Glyph-measured fitting, 11 pt floor, CLASS continuation slides: no overflow, no broken words.

6. **Fast by Design** — vs Slow decks
Parallel diagrams and labs (2 per key), content-hash caches: re-runs are instant.

7. **Safe Generated Code** — vs Unsafe code
SVG whitelist and sanitiser; lab JS passes node --check and may not use network or storage.

VERIFY IT YOURSELF scan / click

Code · Tests · Gold sets · Papers

ADOPTION PATH · low-risk, one step at a time

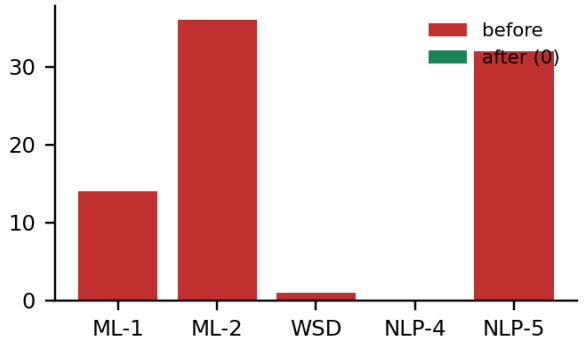
1 · PILOT	2 · DEPT	3 · COLLEGE
Our AI & DS courses; 3–4 faculty	Shared server, deck + lab library	LMS export, regional languages

Learnova

IMPACT AND BENEFITS

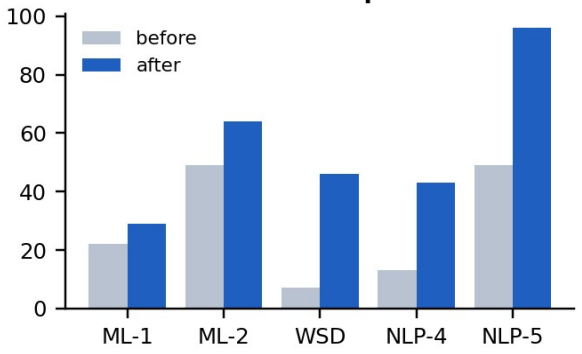
Measured on our own test decks — not on promises

Untitled slides: before → 0



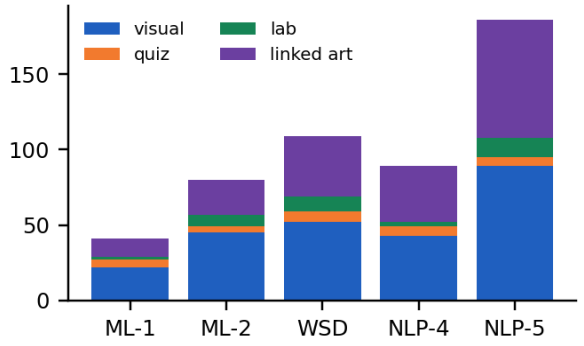
Untitled slides across 5 real decks fell from 83 to 0: every slide now states its point in a full-sentence title.

Slides with a picture



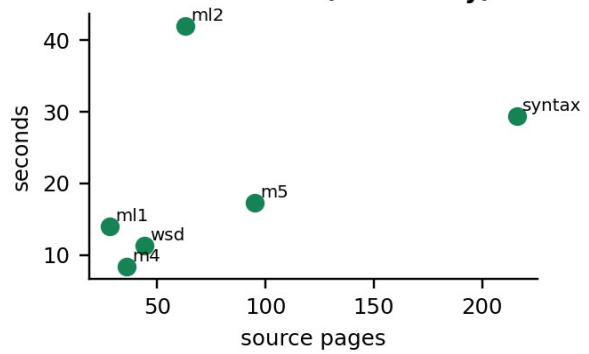
Slides showing a picture rose on every deck — the teacher's figures kept, plus AI diagrams on topic slides.

Added by Learnova (all 0 before)



Visual slides, quizzes and labs went from zero to hundreds, all generated from the deck's own content.

Time to build (no API key)



Run time grows linearly with length: a 216-page PDF becomes a full deck in about 35 s without any API key.

EDUCATIONAL
One idea per slide with a picture; solved steps with reasons; quizzes and labs to practise.

SOCIAL
Works offline and free, so students without paid tools or stable internet get the same material.

INSTITUTIONAL
Consistent, accessible course decks; reusable lab library; supports flipped and remote classes.

TECHNOLOGICAL
Explainable rules, our PSF metric and CLASS splitting; AI used only where it can be checked.

Students
revise with visuals, quizzes, labs

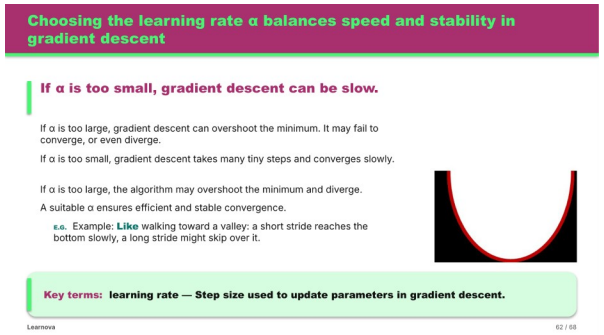
Faculty
old deck redesigned in a minute

HoD / Department
consistent course material

Exam preparation
flashcards, quiz bank, cheat sheet

Remote learners
share link, phone app, offline

Accessibility
readable text, alt text, plain words


Choosing the learning rate α balances speed and stability in gradient descent

If α is too small, gradient descent can be slow.

If α is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.

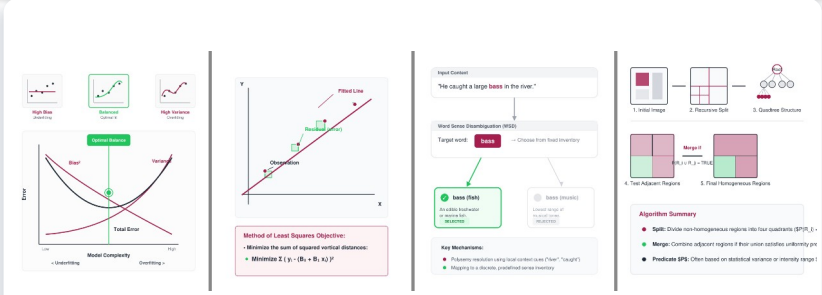
If α is too small, gradient descent takes many tiny steps and converges slowly.

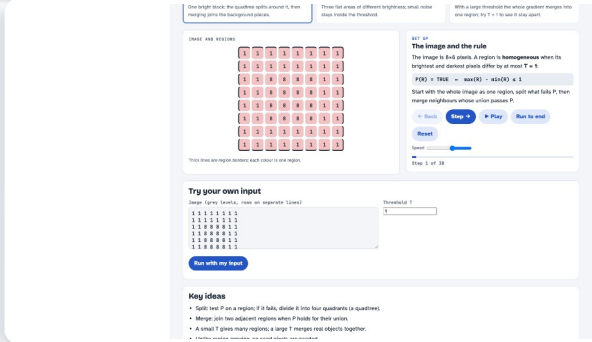
If α is too large, the algorithm may overshoot the minimum and diverge.

A suitable α ensures efficient and stable convergence.

e.g. Example: Like walking toward a valley: a short stride reaches the bottom slowly, a long stride might skip over it.

Key terms: learning rate — Step size used to update parameters in gradient descent.





Learnova

RESEARCH AND REFERENCES

12 road signs · 36 sources · scan any code to check

THE PROBLEM
why old decks fail learners

- **Tufte 2003: Cognitive Style of PowerPoint**
  bullets fragment the argument
  edwardtufte.com
- **Sweller 1988: Cognitive load**
  working memory limits learning
  doi.org/10.1207/s15516709cog1202_4
- **Cowan 2001: the magical number 4**
  about four chunks at a time
  doi.org/10.1017/S0140525X01003922

SLIDE DESIGN
what a good slide looks like

- **Garner & Alley 2013, IJEE**
  assertion-evidence beats bullets
  ijee.ie
- **Alley et al. 2006**
  sentence headlines aid retention
  Technical Communication 53(2)
- **Mayer 2009: Multimedia Learning**
  coherence, signalling, contiguity
  cambridge.org

SLIDE GENERATION
prior systems we improve on

- **Fu et al. 2022: DOC2PPT**
  paper → slides, neural
  doi.org/10.1609/aaai.v36i1.19943
- **Sun et al. 2021: D2S**
  query-based summarisation
  aclanthology.org/2021.naacl-main.111
- **Hu & Wan 2015: PPSGen**
  learned sentence selection
  doi.org/10.1109/TKDE.2014.2359652

FAITHFULNESS
why we keep text verbatim

- **Maynez et al. 2020, ACL**
  summaries hallucinate and omit
  aclanthology.org/2020.acl-main.173
- **Ji et al. 2023, ACM CSUR**
  survey of hallucination
  doi.org/10.1145/3571730
- **Kasneci et al. 2023**
  LLMs in education: risks
  doi.org/10.1016/j.lindif.2023.102274

WORKED EXAMPLES
steps with reasons

- **Sweller & Cooper 1985**
  worked examples teach novices
  doi.org/10.1207/s1532690xci0201_3
- **Atkinson et al. 2000**
  principles of example study
  doi.org/10.3102/00346543070002181
- **Renkl 2014, Cognitive Science**
  example-based learning theory
  doi.org/10.1111/cogs.12086

INTERACTIVE LABS
learning by doing

- **Hundhausen et al. 2002**
  engagement drives visualisation gains
  doi.org/10.1006/jvlc.2002.0237
- **Wieman et al. 2008: PhET**
  simulations that teach
  doi.org/10.1126/science.1161948
- **Chi & Wylie 2014: ICAP**
  interactive > active > passive
  doi.org/10.1080/00461520.2014.965823

LAYOUT & DESIGN RULES
algorithms we adapted

- **Knuth & Plass 1981**
  optimal line breaking → CLASS
  doi.org/10.1002/spe.4380111102
- **Mackinlay 1986: APT**
  automatic visual design rules
  doi.org/10.1145/22949.22950
- **Moritz et al. 2019: Draco**
  design knowledge as constraints
  doi.org/10.1109/TVCG.2018.2865240

DOCUMENTS & OCR
reading real decks

- **Smith 2007: Tesseract OCR**
  open-source OCR engine
  github.com/tesseract-ocr
- **PyMuPDF + LibreOffice**
  PDF geometry, PPTX → PDF
  pymupdf.readthedocs.io
- **python-pptx**
  native, editable PowerPoint
  python-pptx.readthedocs.io

CODE AND TESTS
open, testable, honest

- **Source code**
  755 tests, all modules
  github.com/SDP42/LearnovaV2
- **Gold sets**
  101 labelled slides
  tests/fixtures on GitHub
- **Measurement scripts**
  every number on these slides
  scripts/research on GitHub

TRY IT
run it today

- **Run locally**
  uvicorn API + Vite web app
  README on GitHub
- **Demo decks**
  7 real decks rebuilt
  docs/review on GitHub
- **API docs**
  every endpoint, try it live
  localhost:8000/docs

TOOLS COMPARED
what exists today

- **Gamma**
  prompt → new deck
  gamma.app
- **Microsoft Copilot / Gemini Slides**
  AI inside office suites
  microsoft.com · workspace.google.com
- **Canva Magic Design**
  template-first AI slides
  canva.com

AI MODELS
what powers the AI layer

- **Google Gemini API**
  5-key pool, diagrams, titles
  ai.google.dev